

| National Bio-Engineering Conference 2025<br>Department of Biotechnology and Medical Engineering<br>National Institute of Technology Rourkela |                  |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DATE                                                                                                                                         | TIME             | EVENT                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                   |
| 11th December                                                                                                                                | 9:30 - 10:30 AM  | Inaugration                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 10:30 - 11:00 AM | TEA BREAK                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 11:00 - 12:00 AM | Plenary Talk 1<br>Theme 1: Biomaterials<br>Chair: Dr. Devendra Verma (HOD BM)<br>Co-Chair: Dr. Anju R Babu<br>Dr. Kaushik Chattejee (IISc Bangalore)<br>Topic: Biofabrication<br>Title: Translating 3D-Printed Biomaterials: From the Bench to the Bedside and the Battlefield |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 1:00 - 2:00 PM   | Lunch                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              |                  | Theme 2: Transport Phenomenon<br>Chair: Dr. A Thirugnanam<br>Co-Chair: Dr. Prasoon Kumar                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 2:15-2:45 PM     | Invited Talks 1 (online)<br>Dr. Prashant Agrawal (Northumbria University, UK)<br>Topic: Biofluid Mechanics<br>Title: Simplifying flow of complex fluids in porous media                                                                                                        |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              |                  | Theme 3: Nanomaterials<br>Chair: Dr. Amit Biswas<br>Co-Chair: Dr. Amrita Singh                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 2:45-3:15 PM     | Invited Talks 2 (offline)<br>Dr. Chinmaya Mahapatra (NIT Raipur)<br>Topic: Nanomaterials<br>Title: Emergence of Redox Nanotherapeutics from Quantum Mechanics study to its Cellular to Tissue application                                                                      |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 3:15 - 3:30 PM   | TEA BREAK                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 3:30 - 4:00 PM   | Invited Talks 3 (offline)<br>Dr. Sriparna Chatterjee (IMMT Bhubaneswar)<br>Topic: Nanomaterials<br>Title: Engineering Multifunctional 1D TiO2-based Nanostructures for Targeted Drug Delivery in Cancer Therapy                                                                |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 4:00 - 5:00 PM   | Oral Presentation (Students)                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 12th December    | 9:30 - 10:30 AM                                                                                                                                                                                                                                                                | Plenary Talk 2<br>Theme 4: Plant Molecular Biology<br>Chair: Dr. Angana Sarkar<br>Co-Chair: Dr. Lohit Kumar Srinivas Gujjala<br>Dr. Jitendra Thakur (ICGEB, New Delhi)<br>Topic: Plant Molecular Biology<br>Title: Making Translational Sense of Basic Research: Structure and Function of Plant Mediator Complex |
| 10:30 -10:45 AM                                                                                                                              |                  | Tea Break                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              |                  | Theme 5: Bioinstrumentation and Biomechanics<br>Chair: Bala Chakravarthy Neelapu<br>Co-Chair: Dr. Mirza Khalid Baig                                                                                                                                                            | Theme 6: Micro-Physiological Systems<br>Chair: Dr. Amrita Singh<br>Co-Chair: Dr. Anamika Singh                                                                                                                                                                                                                    |
| 10:45 - 11:15 AM                                                                                                                             |                  | Invited Talks 4 (Offline)<br>Dr. Avinash Eranki (IIT Hyderabad)<br>Topic: Instrumentation<br>Title: Sound Sees, Sound Cures: A Paradigm Shifting Technology for Solid Tumors                                                                                                   | Invited Talks 5 (Offline)<br>Dr. Gowri Balachander (IIT BHU)<br>Topic: Organs-on-a-chip<br>Title: Not Finalised Yet                                                                                                                                                                                               |
| 11:15 - 11: 45 AM                                                                                                                            |                  | Invited Talks 6 (Offline)<br>Dr. Suhail Rizvi (IIT Hyderabad)<br>Topic: Biomechanics<br>Title: From mechanics to metabolism: modeling collagen architecture and ATP dynamics in healthy and diseased articular cartilage                                                       |                                                                                                                                                                                                                                                                                                                   |
| 12:00-1:30 PM                                                                                                                                |                  | LUNCH                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              |                  | Theme 7 : Computational Biology<br>Chair: Dr. Anamika Singh<br>Co-Chair: Dr. Bikash Maharaj                                                                                                                                                                                    | Theme 8: Precision Medicine<br>Chair: Dr. Prasoon Kumar<br>Co-Chair: Dr. Mirza Khalid Baig                                                                                                                                                                                                                        |
| 2:00 - 2:30 PM                                                                                                                               |                  | Invited Talks 7 (Offline)<br>Dr. Abhilash Patel (IIT Kanpur)<br>Topic: System Biology<br>Title: Biomolecular Circuit Design in Synthetic Biology                                                                                                                               | Invited Talks 8 (Online)<br>Dr. Rodriguez Manzano (Imperial College London)<br>Topic: Molecular Diagnostics<br>Title: Not Finalised Yet                                                                                                                                                                           |
| 2:30 -3:00 PM                                                                                                                                |                  | Invited Talks 9 (Offline)<br>Dr. Indresh Yadav (IIT Bhubaneshwar)<br>Topic: Biophysics<br>Title: Engineering at the Molecular Scale: DNA Topology and Its Healthcare Implications                                                                                              | Invited Talks 10 (Online)<br>Dr. Simon C Cork (Anglia Ruskin University, UK)<br>Topic: Obesity<br>Title: Hacking Hunger: The Science behind the future of obesity treatment                                                                                                                                       |
| 3:00 - 3:15 PM                                                                                                                               |                  | Tea Break                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                   |
| 3:15 - 4:15 PM                                                                                                                               |                  | Business Plan Pitching Competition                                                                                                                                                                                                                                             | Oral Presentation (Students)                                                                                                                                                                                                                                                                                      |
| 4:15 - 5:15 PM                                                                                                                               |                  | Scientific Photography Contest                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                   |
| 13th December                                                                                                                                |                  | Theme: Bioentrepreneurship<br>Chair: Dr. Prasoon Kumar<br>Co-Chair: Dr. Lohit Kumar Srinivas Gujjala                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 9:30 - 10:00 AM  | Invited Talks 11 (Offline)<br>Dr. Kaushik Choudhury (AIC-IPR)<br>Topic: Bioentrepreneurship<br>Title: Ctrl + Alt + Disrupt: Hacking the Start-up Game as an Engineer                                                                                                           |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 10:00 - 11:30 AM | Poster Presentation and Tea Break (CHAI PE CHARCHA)                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                   |
|                                                                                                                                              | 11:30 AM         | Valedictory Function                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                   |